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# Deep Learning

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## Problem

- Automatic understanding of human behavior from visual data remains a challenging task in computer vision due to variations in lighting, pose, background , and subject appearance.
- Emotion recognition from facial images and human activity recognition from visual data are two important problems with applications in human-computer interaction, surveillance, healthcare, and intelligent systems.



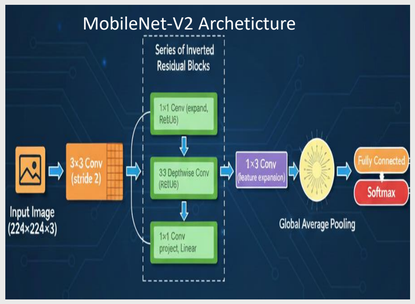

## Goal

- The goal of this project is to systematically evaluate and compare deep convolutional neural networks for two visual recognition tasks:
- Facial Emotion Recognition
- Human Activity Recognition
- The study aims to analyze performance, efficiency, and generalization of both custom-built and pretrained CNN models under a unified experimental setup.



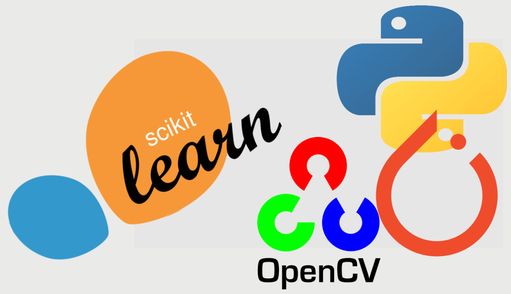

## Approach

- Datasets: FER-2013, UCF101
- A custom CNN was designed and trained from scratch as a baseline
- Five pretrained CNN models (ResNet-18, ResNet-50, MobileNet-V2, EfficientNet-B0, VGG-16) were fine-tuned using transfer learning.
- Identical preprocessing, data loaders, and training settings were applied to ensure fair comparison.




## Technologies used

- Programming Language: Python
- Deep Learning Framework: PyTorch
- Pretrained Models: Torchvision
- Data Processing: OpenCV, NumPy, Pandas
- Evaluation: Scikit-learn




## Results and Conclusion

• Efficiency Winner: MobileNet-V2 offers the best accuracy-to-size ratio across both tasks, while VGG-16 (134M params) underperformed on activity despite being the largest model.

• Transfer Learning Impact: ImageNet pretraining enabled rapid convergence (10-15 epochs) and consistently outperformed custom CNN baselines in both accuracy and training speed.

• Production Recommendation: Deploy MobileNet-V2 for both tasks due to optimal balance of performance (82% activity, 66% emotion), efficiency (2.2M params), and fast inference.

• Future Improvements: Ensemble methods, advanced regularization (dropout, mixup), and task-specific augmentation could boost accuracy by 2-5% while reducing overfitting gaps.

